

Regulated Pure Pursuit for Robot Path Tracking

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Abstract

The accelerated deployment of service robots have spawned a number of algorithm variations to better handle real-world conditions. Many local trajectory planning techniques have been deployed on practical robot systems successfully. While most formulations of Dynamic Window Approach and Model Predictive Control can progress along paths and optimize for additional criteria, the use of pure path tracking algorithms is still commonplace. Decades later, Pure Pursuit and its variants continues to be one of the most commonly utilized classes of local trajectory planners. However, few Pure Pursuit variants have been proposed with schema for variable linear velocities - they either assume a constant velocity or fails to address the point at all. This paper presents a variant of Pure Pursuit designed with additional heuristics to regulate linear velocities, built atop the existing Adaptive variant. The *Regulated Pure Pursuit algorithm* makes incremental improvements on state of the art by adjusting linear velocities with particular focus on safety in constrained and partially observable spaces commonly negotiated by deployed robots. We present experiments with the Regulated Pure Pursuit algorithm on industrial-grade service robots. We also provide a high-quality reference implementation that is freely included ROS 2 Nav2 framework at <https://github.com/ros-planning/navigation2> for fast evaluation.

Keywords: Service Robots, Mobile Robots, Motion Planning, Path Planning

1 Introduction

Dynamic Window Approach (DWA) [1], Pure Pursuit (PP) [6], and Model Predictive Control (MPC) [2] are by far the most commonly deployed path trackers. They all have a strong heritage for reliability in a wide range of environmental conditions. DWA and MPC are often, but not

always, formulated as multi-objective trajectory generation problems to maximize criteria such as avoiding dynamic obstacle collisions on top of path tracking. This has made them particularly well suited for many robotics applications where dynamic robot behaviors are rewarded. A great deal of work has been conducted on these allowing

对机器人路径跟踪的监管纯粹追求

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Abstract

服务机器人的加速部署催生了许多算法变体，以更好地处理现实世界的情况。许多局部轨迹规划技术已成功部署在实际的机器人系统上。虽然动态窗口方法和模型预测控制的大多数公式可以沿着路径前进并针对附加标准进行优化，但纯路径跟踪算法的使用仍然很常见。几十年后，Pure Pursuit 及其变体仍然是最常用的局部轨迹规划器类别之一。然而，很少有人提出具有可变线速度模式的纯粹追踪变体——它们要么假设速度恒定，要么根本无法解决这一点。本文提出了 Pure Pursuit 的一个变体，该变体设计有额外的启发式方法来调节线性速度，构建在现有的 Adaptive 变体之上。

Regulated Pure Pursuit algorithm 通过调整线速度来逐步改进现有技术，特别关注通常由部署的机器人协商的受限和部分可观察空间中的安全性。我们展示了在工业级服务机器人上使用受监管的纯追踪算法进行的实验。我们还提供了免费包含在 ROS 2 Nav2 框架中的高质量参考实现（位于 <https://github.com/ros-planning/navigation2>），以便快速评估。

Keywords: 服务机器人、移动机器人、运动规划、路径规划

1 Introduction

动态窗口方法 (DWA) [1]、纯追踪 (PP) [6] 和模型预测控制 (MPC) [2] 是迄今为止最常部署的路径跟踪器。它们都具有在各种环境条件下都具有可靠性的悠久传统。DWA 和 MPC 经常使用，但不是

始终被表述为多目标轨迹生成问题，以最大化标准，例如在路径跟踪之上避免动态障碍物碰撞。这使得它们特别适合许多奖励动态机器人行为的机器人应用。在这些方面已经开展了大量的工作

them to be found on many commercially available robots today.

However, there still exist many applications of deployed robot systems where this can be considered a detractor. Surveyed among high-end research and service robot navigation systems, pure path tracking continues to be a common theme in many robots [3, 4]. Among single-objective path trackers, a simple and reliable method continues to be exploited decades after its development: Pure Pursuit.

Pure Pursuit uses simple geometry to find the curvature of a path required to drive a robot towards a given point on the path. The algorithm itself does not place any stability restrictions on the translational velocities during operation, however it also lacks any schema for selecting them. Near-universally, implementations use a fixed speed. Many variations of Pure Pursuit exist, however most address the more obvious area of the selection of lookahead points, which largely aids in stabilizing convergence behaviors towards the path at a wider range of velocities. The Pure Pursuit algorithm was not developed with service and industrial robots in mind, which have additional safety requirements making it further unrealistic to move at a fixed velocity.

This work proposes an incremental improvement on the Pure Pursuit path tracking algorithm by describing a reference method of adjusting translational and rotational velocities to improve safety and operability in a broad range of common deployed robot applications. We improve the Adaptive Pure Pursuit (APP) algorithm [7] by regulating velocities via penalizing sharp changes in path curvature and proximity to obstacles - two of the most common events requiring conscientious navigation behaviors. The *Regulated Pure Pursuit* (RPP) algorithm slows sharp turns into partially observable dynamic environments (aisles, hallways, intersections) to reduce the likelihood and impact of potential collisions. It also reduces linear velocity in close proximity to obstacles such as people and fixed infrastructure to reduce the likelihood of collision in constrained indoor environments. Finally, it additionally includes preemptive collision detection missing from other variations.

Another main contribution of this work is in describing and providing free and high quality implementations of PP, APP, and RPP for evaluation. It is integrated into the Nav2 mobile robot

navigation system commonly used by researchers and adds a new capability to the framework [5]. This work is well documented, tested, and is in use on robots deployed today.

This paper is organized as follows. First, in section 2, we will describe the conventional Pure Pursuit Algorithm, providing a mathematical formulation that will allow us to describe related works that provide variations to the original algorithm. In section 3, we describe the Regulated Pure Pursuit Algorithm, our main contribution to this paper. Section 4 will describe its implementation in a real robotic system, and in section 5, we will experimentally validate the improvement introduced with our contribution. Finally, we will provide the conclusions in section 7.

2 Related Work

Pure Pursuit [7] is a widely used algorithm for path tracking. It is simple but effective. Considering a path $\mathcal{P}$ as an ordered list of points $\mathcal{P} = \{p_0, p_1, \dots, p_n\}$ where $p_i = (x_i, y_i) \in \mathcal{P}$. A local trajectory planner is a function f that determines the linear and angular velocity to track a reference path $\mathcal{P}_t$ at time t (Eq. 1).

$$(v_t, \omega_t) = f(\mathcal{P}_t) \quad (1)$$

Figure 1 shows visual geometric representations of Pure Pursuit. First, it determines the closest point p_r on $\mathcal{P}_t$ to the robot position. Using a given lookahead distance, L , the lookahead point p_l is determined as the first p_i at least L distance away from p_r in Eq. 2.

$$\begin{aligned} dist(p_i) &= \sqrt{(x_r - x_i)^2 + (y_r - y_i)^2} \\ p_l &= p_i \in \mathcal{P}_t, \begin{cases} dist(p_{i-1}) < L \\ dist(p_i) \geq L \end{cases} \end{aligned} \quad (2)$$

With a known p_l , we can determine curvature of the circle (recall, $R = 1/\kappa$) using simple geometry. If $\mathcal{P}_t$ is represented in vehicle base coordinates, $\mathcal{P}'_t$, where the robot position is the origin, then the curvature can be represented as

$$\kappa = \frac{2 y'_l}{L^2} \quad (3)$$

如今，在许多商用机器人上都可以找到它们。

然而，仍然存在许多已部署的机器人系统的应用，这可以被视为一个缺点。在高端研究和服务机器人导航系统中进行的调查发现，纯路径跟踪仍然是许多机器人的共同主题 [3, 4]。在单目标路径跟踪器中，一种简单而可靠的方法在其开发数十年后继续被开发：Pure Pursuit。

Pure Pursuit 使用简单的几何形状来找到驱动机器人朝向路径上给定点所需的路径曲率。该算法本身在操作过程中不会对平移速度施加任何稳定性限制，但它也缺乏任何选择它们的模式。几乎普遍地，实现都使用固定速度。纯粹追踪存在许多变体，但是大多数都解决了选择前瞻点的更明显的区域，这在很大程度上有助于稳定在更广泛的速度下朝向路径的收敛行为。Pure Pursuit 算法的开发并未考虑到服务和工业机器人，这些机器人有额外的安全要求，使得以固定速度移动更加不现实。

这项工作通过描述调整平移和旋转速度的参考方法，提出了对 Pure Pursuit 路径跟踪算法的增量改进，以提高广泛的常见部署机器人应用中的安全性和可操作性。我们改进了自适应纯粹追踪 (APP) 算法 [7]，通过惩罚路径曲率和接近障碍物的急剧变化（这是需要认真导航行为的两个最常见事件）来调节速度。Regulated Pure Pursuit (RPP) 算法会减慢急转弯速度，进入部分可观察的动态环境（过道、走廊、十字路口），以减少潜在碰撞的可能性和影响。它还降低了靠近障碍物（例如人和固定基础设施）的线速度，以减少在受限室内环境中发生碰撞的可能性。最后，它还包括其他变体中缺少的先发制人的碰撞检测。

这项工作的另一个主要贡献是描述和提供免费且高质量的 PP、APP 和 RPP 实现以供评估。它集成到 Nav2 移动机器人中

研究人员常用的导航系统，并为框架添加了新的功能 [5]。这项工作有详细记录、经过测试，并已在当今部署的机器人上使用。

本文的结构如下。首先，在第 2 节中，我们将描述传统的纯追踪算法，提供一种数学公式，使我们能够描述为原始算法提供变体的相关工作。在第 3 节中，我们描述了调节纯追踪算法，这是我们对本文的主要贡献。第 4 节将描述其在真实机器人系统中的实现，在第 5 节中，我们将通过实验验证我们的贡献所带来的改进。最后，我们将在第 7 节中提供结论。

2 Related Work

Pure Pursuit [7] 是一种广泛使用的路径跟踪算法。它简单但有效。将路径 $\mathcal{P}$ 视为点 $\mathcal{P} = \{p_0, p_1, \dots, p_n\}$ （其中 $p_i = (x_i, y_i) \in \mathcal{P}$ ）的有序列表。局部轨迹规划器是一个函数 f ，它确定在时间 t （时跟踪参考路径 $\mathcal{P}_t$ 的线速度和角速度。1）。

$$(v_t, \omega_t) = f(\mathcal{P}_t) \quad (1)$$

图 1 显示了 Pure Pursuit 的视觉几何表示。首先，它确定 $\mathcal{P}_t$ 上距离机器人位置最近的点 p_r 。使用给定的先行距离 L ，先行点 p_l 被确定为距离 p_r 至少 L 距离的第一个 p_i 。2。

$$\begin{aligned} dist(p_i) &= \sqrt{(x_r - x_i)^2 + (y_r - y_i)^2} \\ p_l &= p_i \in \mathcal{P}_t, \begin{cases} dist(p_{i-1}) < L \\ dist(p_i) \geq L \end{cases} \end{aligned} \quad (2)$$

有了已知的 p_l ，我们就可以使用简单的几何来确定圆的曲率（回想一下， $R = 1/\kappa$ ）。如果 $\mathcal{P}_t$ 用车辆基坐标 $\mathcal{P}'_t$ 表示，其中机器人位置是原点，则曲率可以表示为

$$\kappa = \frac{2 y'_l}{L^2} \quad (3)$$

where κ is the path curvature required to drive the robot from its starting position to the lookahead point, y'_l is the lateral coordinate of the lookahead point p'_l , and L is the desired distance between p_r and p_l .

Figure 1 shows this visually, where L is represented geometrically as the circle's chord. With curvature to drive towards the lookahead point, the commands can be sent to a robot controller. This process is then updated at the desired rate.

The primary parameters of the Pure Pursuit path tracker are simply the velocity of travel v_t and the distance L along the path used to select the lookahead point p_l . In the standard formulation, this lookahead point p_l is a distance L from the robot tuned to achieve an acceptable trade-off between oscillations centered around the path (shorter distances) and slower convergence (longer distances). There exists a broad range of admissible lookahead distances [6].

PP and its variants are applicable to Ackermann and differential-drive robots due to PP's formulation supporting dynamics with longitudinal motion and a turning rate in body-fixed frame. For Ackermann robots to use PP, the path must be feasible given the kinematic-constraints of the platform - while differential-drive robots may follow any holonomic path using PP. While omni-directional robots may utilize PP variants, they will be restricted from performing lateral movements.

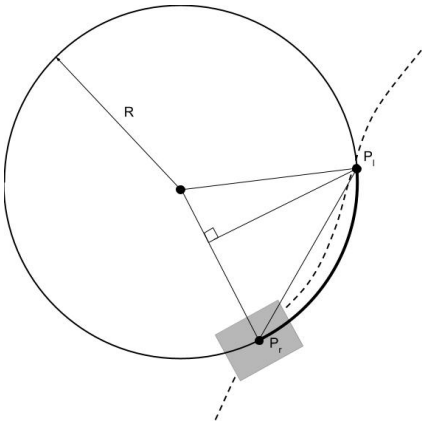


Fig. 1 Geometry of finding the path curvature.

There are several known downsides of this approach. In high curvature situations, Pure Pursuit is known to have overshoot or undershoot

behaviors resulting in path deviations, even in a well tuned system [6]. This is typically not a major concern for autonomous driving applications which naturally has a minimum turning radius limit, but a more substantive issue for smaller-scale applications like industrial and consumer robots. It also does not specify any translational velocity specification criteria during execution. While this can be beneficial as it allows for a great deal of flexibility with different linear velocity profiles, in practice, without described methods, nearly all known variants use a constant translational velocity profile. This is an unsafe defacto-standard.

Variants of this algorithm have been proposed to increase path tracking stability by varying computations of the lookahead point. MIT's entry into the DARPA Urban Challenge implemented the *Adaptive Pure Pursuit* (APP) algorithm for lane following while varying the lookahead distances proportionally to the translational velocity [7]. For velocities in the operating range, a mapping of lookahead distances to velocities is required such that they have an acceptable trade-off between oscillation and slower convergence to the path. A common formulation for this is in Eq. 4, where L_t is the lookahead distance, v_t is the translational velocity, and l_t is a lookahead gain representing the amount of time to project v_t forward [8].

$$L_t = v_t l_t \quad (4)$$

A recent variant has been created to significantly improve path tracking accuracy of Pure Pursuit [?]. This variant also modifies the process of determining the lookahead point, but instead adjusts this point off the path to address the long-standing edge case of short-cutting during path tracking in substantial turns. While it also contains a heuristic change to the translational velocity during that edge-case, its policy is derived from on-road properties and ackermann kinematics that does not generalize to other applications outside of on-road driving of Ackermann vehicles. It does not solve the problems facing deployment of PP techniques to general mobile robotics applications for which this work focuses (e.g. practicable translational velocity control); though its improvements in path tracking accuracies are notable.

其中 κ 是驱动机器人从起始位置到前瞻胡萝卜所需的路径曲率, y_l 是前瞻点 p_l 的横向坐标, L 是 p_r 和 p_l 之间的所需距离。

图 1 直观地展示了这一点, 其中 L 在几何上表示为圆的弦。通过曲率驱动前瞻点, 命令可以发送到机器人控制器。然后以所需的速率更新该过程。

Pure Pursuit 路径跟踪器的主要参数只是行进速度 v_t 和沿用于选择前瞻点 p_l 的路径的距离 L 。在标准公式中, 该前瞻点 p_l 是距机器人的距离 L , 该距离经过调整以在以路径为中心的振荡 (较短距离) 和较慢的收敛 (较长距离) 之间实现可接受的权衡。允许的前瞻距离范围很广 [6]。

PP 及其变体适用于阿克曼和差动驱动机器人, 因为 PP 的公式支持车身固定框架中的纵向运动和转动速率动力学。对于使用 PP 的阿克曼机器人来说, 考虑到平台的运动学约束, 该路径必须是可行的 - 而差动驱动机器人可以使用 PP 遵循任何完整路径。虽然全向机器人可以使用 PP 变体, 但它们的横向运动将受到限制。

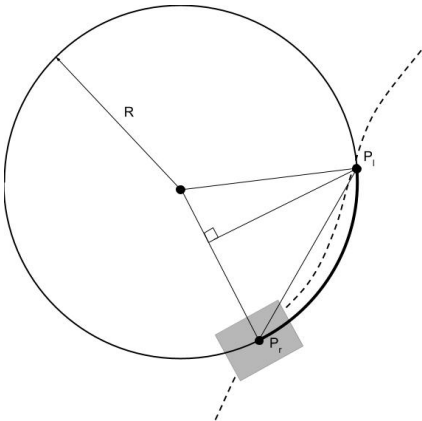


Fig. 1 寻找路径曲率的几何。

这种方法有几个已知的缺点。在高曲率情况下, Pure Pursuit 会出现过冲或下冲的情况

即使在经过良好调整的系统, 行为也会导致路径偏差 [6]。对于自动驾驶应用来说, 这通常不是一个主要问题, 因为自动驾驶应用自然有最小转弯半径限制, 但对于工业和消费机器人等较小规模的应用来说, 这是一个更实质性的问题。它在执行期间也没有指定任何平移速度规范标准。虽然这可能是有益的, 因为它允许不同线速度分布的很大灵活性, 但实际上, 在没有描述的方法的情况下, 几乎所有已知的变体都使用恒定的平移速度分布。这是一个不安全的事实上的标准。

人们已经提出了该算法的变体, 通过改变前瞻点的计算来提高路径跟踪稳定性。麻省理工学院参加 DARPA 城市挑战赛实施了用于车道跟踪的 Adaptive Pure Pursuit (APP) 算法, 同时根据平移速度按比例改变前视距离 [7]。对于工作范围内的速度, 需要将前瞻距离映射到速度, 以便它们在振荡和较慢收敛到路径之间具有可接受的权衡。一个常见的公式是方程 4, 其中 L_t 是前瞻距离, v_t 是平移速度, l_t 是前瞻增益, 表示向前投射 v_t 的时间量 [8]。

$$L_t = v_t l_t \quad (4)$$

最近创建的一个变体可以显著提高 Pure Pursuit [?] 的路径跟踪精度。该变体还修改了确定先行点的过程, 但是将该点调整为偏离路径, 以解决在大量转弯的路径跟踪期间长期存在的短路边缘情况。虽然它还包含对边缘情况下平移速度的启发式变化, 但其策略源自道路属性和阿克曼运动学, 不能推广到阿克曼车辆道路驾驶之外的其他应用。它没有解决将 PP 技术部署到本工作重点关注的通用移动机器人应用中所面临的问题 (例如实用的平移速度控制); 尽管它在路径跟踪精度方面的改进是显著的。

Pure Pursuit nor its variations account for dynamic effects of the vehicle. As PP is geometrically derived, vehicle dynamics are not modeled in this style of path tracking algorithm. There are other types of path trackers that do account for vehicle dynamics [9]. This paper uses PP and its variants for comparison to offer a candid contrast with the same or directly analog parameters to clearly indicate the differences in performance due to the contribution ¹.

3 Regulated Pure Pursuit

The Regulated Pure Pursuit algorithm is designed for service and industrial mobile robots in real-world constrained and partially observable environments. It provides methods for adapting the robot's translational velocity to current conditions; robot systems cannot simply barrel through aisles and facilities at full speed without regard. These methods are linear regulation cost functions that provide high-quality behavior across a variety of practical mobile robot environments derived from commonplace requirements on reducing robot velocities in the presence of sharp turns or when operating in confined regions.

The first phase of the Regulated Pure Pursuit algorithm is to transform the input path $\mathcal{P}_t$ $\{p_0, \dots, p_j, p_r, \dots, p_n\}$ into the robot's base coordinate frame $\mathcal{P}'_t$ and prune it. In doing so, determining the curvature of the path is reduced to the simple algebraic expression shown in Eq. 3. Before transformation, p_r is determined and all prior points $\{p_0, \dots, p_j\}$ are permanently pruned from the stored path to prevent unnecessary future transformation of obsolete data. The transformed path $(\{p'_r, \dots, p'_n\})$ is also pruned for all p_i points where $\text{dist}(p'_r, p'_i) \gg L_t$ as they are sufficiently far away that they will never need to be considered at t . These far path points continue to be held in the stored path for future iterations, until a new path is received, as the robot progresses along the path.

Next, RPP will utilize the same lookahead selection mechanics as Adaptive Pure Pursuit described in Eq. 4. The lookahead distance L_t is

thusly proportional to the speed v_t and a lookahead gain l_t such that longer distances are used while moving faster. This stabilizes the path tracking behavior over larger ranges of translational velocities [10]. This distance is used to select the lookahead point p_l . While interpolating between path points was found to demonstrably improve smoothness on sparse paths at autonomous vehicle speeds, empirically this did not contribute much benefit at the service and industrial robot speeds using typical grid map planning resolutions (0.025 m - 0.1 m) [8]. However, interpolation is beneficial for use in sparser path resolutions (0.1 - 1.0m).

The desired linear velocity, v_t , is next further processed by the curvature and proximity heuristics. Both heuristics are applied to linear velocity and we take the maximum of the two.

The purpose of the **curvature heuristic** is to slow the robot to v'_t during sharp turns into partially observable environments, such as when entering or exiting hallways and aisles commonly found in retail, warehouses, factories, schools, and shopping malls. This allows for significantly safer traversal when making blind turns. This heuristic is applied to the linear velocity v_t when the change in curvature θ is above a minimum threshold T_θ . This minimum radius restricts velocity scaling in minor turns or path variations that do not require slowing. The curvature velocity v'_t returned by this heuristic is selected as:

$$v'_t = \begin{cases} v_t & \kappa > T_\kappa, \\ \frac{v_t}{r_{min} \kappa} & \kappa \leq T_\kappa \end{cases} \quad (5)$$

Where r_{min} is the minimum radius to apply the heuristic. This formulation is a mathematical reduction of a simple error calculation between the minimum radius and the radius of a circle represented by κ and can be trivially derived.

The **proximity heuristic** is applied to the linear velocity v_t when the robot becomes in close proximity to dynamic obstacles or fixed infrastructure. The purpose of this is to slow the robot when in constrained environments where the potential for collision is particularly high. Reducing the speed near fixed infrastructure lowers the likelihood of collision by lessening the impact of small path variations in tight spaces. Lowering the speed of industrial and service robots in close proximity to dynamic agents, such as humans, is a common safety requirement- allowing a robot to reactively

¹Not leaving the opportunity for misleading results by comparing unrelated families of methods with unknown degrees of tuning

纯粹的追求及其变体都考虑了车辆的动态效果。由于 PP 是几何推导的，因此车辆动力学不是用这种路径跟踪算法建模的。还有其他类型的路径跟踪器确实考虑了车辆动力学 [9]。本文使用 PP 及其变体进行比较，以与相同或直接模拟的参数进行坦率对比，以清楚地表明由于贡献¹而导致的性能差异。

3 Regulated Pure Pursuit

Regulated Pure Pursuit 算法专为现实世界受限和部分可观察环境中的服务和工业移动机器人而设计。它提供了使机器人的平移速度适应当前条件的方法；机器人系统不能不顾一切地全速穿过过道和设施。这些方法是线性调节成本函数，可在各种实际移动机器人环境中提供高质量的行为，这些环境源自在急转弯或在有限区域内操作时降低机器人速度的常见要求。

调节纯追踪算法的第一阶段是将输入路径 $\mathcal{P}_t$ $\{p_0, \dots, p_j, p_r, \dots, p_n\}$ 转换为机器人的基本坐标系 $\mathcal{P}'_t$ 并对其进行修剪。这样做时，路径曲率的确定被简化为方程式所示的简单代数表达式。3. 在转换之前，确定 p_r ，并从存储的路径中永久删除所有先前的点 $\{p_0, \dots, p_j\}$ ，以防止将来对过时数据进行不必要的转换。变换后的路径 $(\{p'_r, \dots, p'_n\})$ 也会针对 $\text{dist}(p'_r, p'_i) \gg L_t$ 处的所有 p_i 点进行修剪，因为它们足够远，因此永远不需要在 t 处考虑它们。这些远路径点继续保留在存储的路径中以供将来迭代，直到随着机器人沿着该路径前进而接收到新路径。

接下来，RPP 将利用与等式 1 中描述的自适应纯追踪相同的前瞻选择机制。4. 前瞻距离 L_t 为

因此与速度 v_t 和前瞻增益 l_t 成正比，使得在移动更快的同时使用更长的距离。这可以稳定较大平移速度范围内的路径跟踪行为[10]。该距离用于选择先行点 p_i 。虽然发现路径点之间的插值可以明显提高自动驾驶车辆速度下稀疏路径的平滑度，但从经验来看，使用典型的网格地图规划分辨率 (0.025 m - 0.1 m) [8]，这对服务和工业机器人速度没有带来太大好处。然而，插值有利于在稀疏路径分辨率 (0.1 - 1.0m) 中使用。

接下来通过曲率和邻近启发法进一步处理所需的线速度 v_t 。两种启发法都应用于线速度，我们取两者中的最大值。

curvature heuristic 的目的是在急转弯进入部分可观察环境时将机器人减速至 v'_t ，例如在进入或退出零售、仓库、工厂、学校和购物中心常见的走廊和过道时。这使得盲转弯时的行驶更加安全。当曲率变化 θ 高于最小阈值 T_θ 时，该启发式应用于线速度 v_t 。此最小半径限制了不需要减速的小转弯或路径变化的速度缩放。该启发式返回的曲率速度 v'_t 选择为：

$$v'_t = \begin{cases} v_t & \kappa > T_\kappa, \\ \frac{v_t}{r_{min} \kappa} & \kappa \leq T_\kappa \end{cases} \quad (5)$$

其中 r_{min} 是应用启发式的最小半径。该公式是最小半径和由 κ 表示的圆的半径之间的简单误差计算的数学简化，并且可以简单地导出。

当机器人非常接近动态障碍物或固定基础设施时，**proximity heuristic** 应用于线速度 v_t 。这样做的目的是在碰撞可能性特别高的受限环境中减慢机器人速度。降低固定基础设施附近的速度可以减少狭小空间内小路径变化的影响，从而降低碰撞的可能性。降低工业和服务机器人靠近动态代理（例如人类）的速度是一项常见的安全要求，允许机器人做出反应

¹Not leaving the opportunity for misleading results by comparing unrelated families of methods with unknown degrees of tuning

stop faster to prevent potential injury. The linear formulation of this heuristic reduces the speed by the ratio of d_O/d_{prox} with a gain, α , to adjust the response for individual systems. The other formulations tested, such as exponential and quadratic, far too significantly penalized proximity to objects and contained only a narrow band of gains which would result in an acceptable trade-off between proximity to obstacles and velocities to accomplish the robotic task. This linear heuristic in Eq. 6 has a broad range of α which may be finely tuned by a system designer and is derivative of the well-used Adaptive Pure Pursuit’s formulation [10].

$$v'_t = \begin{cases} v_t \frac{\alpha d_O}{d_{prox}} & d_O \leq d_{prox} \\ v_t & d_O > d_{prox} \end{cases} \quad (6)$$

Where d_{prox} is the proximity distance to obstacles to apply the heuristic, d_O is the current distance to an obstacle, and α is a gain to scale the heuristic function for aggressive behavior, with the requirement that $\alpha \leq 1.0$. A higher α lowers the velocity of the robot in proximity to obstacles more expeditiously. The value of d_{prox} should be established based on the system requirements of a robot’s application for how close an obstacle can be before the robot begins slowing its maximum velocity.

After velocity regulation, the algorithm then determines the path curvature using Eq. 1. The angular velocity is computed using the regulated velocity, not desired linear velocity, which prevents consequential undershoot behavior relative to target curvatures [6]. Finally, the angular velocity, ω_t , is then trivially found as Eq. 7.

$$\omega_t = v'_t \kappa \quad (7)$$

The final step of the algorithm is to check our path tracking command for current or imminent collisions, new to RPP. A given angular velocity ω_t and regulated linear velocity v'_t can be projected forward in time, resulting in a circular arc. Points on the arc are sampled at the grid map cell resolution forward for a set duration. Collision checking is done based on a duration to collision rather to the lookahead point such that the robot is always, at minimum, a set duration from collision. At slow speeds, it may not be sensible to collision check to a lookahead point tens of meters, or hundreds of seconds, away. Rather, a temporal schema allows

for fine maneuvers in confined spaces where the current velocity commands may not be admissible a short distance - but long time - away.

This controller is known to converge from the stability analysis completed in [11]. Ollera’s work on the general Pure Pursuit algorithm applies to Regulated Pure Pursuit as well, showing that straight paths and constant curvature paths are stable. Pure Pursuit itself is independent of velocity in its formulation. It is only used in the final step to convert the path curvature into a rotational velocity for a base to track. The regulation heuristics proposed in this variant do not change the basis of stability of the root algorithm.

4 Implementation

Another contribution of this work is a high-quality reference implementation of Regulated Pure Pursuit. This reference implementation is available as one of the default path tracking algorithm plugins in the new and improved ROS 2 Navigation System, known as Nav2. Nav2 is a scalable navigation framework with multiple algorithm implementations, documentation, and support for building modern and reliable research and commercial navigation systems [5]. This optimized C++ implementation has 92% unit test coverage and is used in the experiments in Section 5. It also contained a few additional features that are highlighted in this section.

Rather than creating distinct implementations of Pure Pursuit, Adaptive Pure Pursuit, and Regulated Pure Pursuit, all three have been built into a single reference implementation parameterized to enable each specific behavior. This allows us to test these algorithms easily and analyze their run-time performance more closely knowing that they share the vast majority of their computations. This also allows researchers and developers to quickly evaluate and tune these features by simply changing a handful of parameters from the available binaries.

Additionally, there is a setting to allow the robot to slow its speed when approaching the target goal pose. This feature allows the robot to come to a more gradual stop. When the robot is within close proximity to the goal, the translational velocity is lowered proportional to the remaining distance, up to a minimum viable velocity to make progress.

更快地停车以防止潜在的伤害。该启发式的线性公式通过 d_O/d_{prox} 与增益 α 的比率降低速度，以调整各个系统的响应。测试的其他公式，例如指数公式和二次公式，对接近物体的影响太大，并且仅包含狭窄的增益范围，这将导致在接近障碍物和完成机器人任务的速度之间取得可接受的权衡。方程中的线性启发式。6 具有广泛的 α 范围，系统设计人员可以对其进行微调，并且是广泛使用的 Adaptive Pure Pursuit 公式的派生物 [10]。

$$v'_t = \begin{cases} v_t \frac{\alpha d_O}{d_{prox}} & d_O \leq d_{prox} \\ v_t & d_O > d_{prox} \end{cases} \quad (6)$$

其中， d_{prox} 是应用启发式的到障碍物的接近距离， d_O 是到障碍物的当前距离， α 是针对攻击行为缩放启发式函数的增益，要求 $\alpha \leq 1.0$ 。 α 越高，机器人接近障碍物的速度就越快。 d_{prox} 的值应根据机器人应用的系统要求来确定，即机器人开始减慢其最大速度之前障碍物的距离有多近。

速度调节后，算法然后使用方程式确定路径曲率。1. 角速度是使用调节速度而不是期望的线速度来计算的，这可以防止相对于目标曲率的相应下冲行为[6]。最后，角速度 ω_t 可以简单地找到，如下式：7。

$$\omega_t = v'_t \kappa \quad (7)$$

该算法的最后一步是检查我们的路径跟踪命令是否存在当前或即将发生的碰撞，这对于 RPP 来说是新的。给定的角速度 ω_t 和调节的线速度 v'_t 可以在时间上向前投影，从而产生圆弧。弧上的点在设定的持续时间内以网格图单元分辨率向前采样。碰撞检查是根据碰撞持续时间而不是前瞻点来完成的，以便机器人始终至少在设定的碰撞持续时间内进行。在低速情况下，对数十米或数百秒外的前瞻点进行碰撞检查可能并不明智。相反，时间模式允许

用于在有限空间内进行精细机动，在这些空间中，当前速度命令可能无法在短距离（但长时间）之外使用。

从[11]中完成的稳定性分析可知，该控制器会收敛。Ollera 在通用 Pure Pursuit 算法上的工作也适用于 Regulated Pure Pursuit，表明直线路径和恒定曲率路径是稳定的。Pure Pursuit 本身的配方与速度无关。它仅在最后一步中使用，将路径曲率转换为基座跟踪的旋转速度。该变体中提出的调节启发式不会改变根算法稳定性的基础。

4 Implementation

这项工作的另一个贡献是 Regulated Pure Pursuit 的高质量参考实现。该参考实现可作为新的和改进的 ROS 2 导航系统（称为 Nav2）中的默认路径跟踪算法插件之一。Nav2 是一个可扩展的导航框架，具有多种算法实现、文档并支持构建现代可靠的研究和商业导航系统[5]。这种优化的 C++ 实现具有 92% 的单元测试覆盖率，并在第 5 节的实验中使用。它还包含本节中重点介绍的一些附加功能。

不是创建 Pure Pursuit、Adaptive Pure Pursuit 和 Regulated Pure Pursuit 的不同实现，而是将所有这三个实现构建到参数化的单个参考实现中以启用每种特定行为。这使我们能够轻松测试这些算法并更仔细地分析它们的运行时性能，因为我们知道它们共享绝大多数计算。这还允许研究人员和开发人员通过简单地更改可用二进制文件中的一些参数来快速评估和调整这些功能。

此外，还有一个设置允许机器人在接近目标姿势时减慢速度。此功能允许机器人更加逐渐地停止。当机器人非常接近目标时，平移速度会与剩余距离成比例地降低，直到取得进展的最小可行速度。

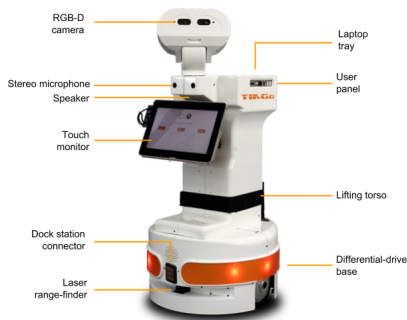


Fig. 2 The Pal Robotics Tiago used in the experiments [12].

This work is specifically targeted at industrial and service robots, many of which are differential drive. While this is of no technical concern to Pure Pursuit, many differential drive robots will utilize holonomic search-based path planners (such as Dijkstra’s or A*) rather than kinematically feasible planners frequently used by Ackermann vehicles [13]. This is of concern to the Regulated Pure Pursuit implementation as the relative heading of the path may not align with the robot’s starting heading. In this case, it is beneficial for the robot to be able to rotate to a rough starting heading before beginning to track the holonomic path. Further, rotating to the final heading when the robot is within the translational goal tolerances is useful when being deployed in orientation-sensitive applications. These are non-issues when using feasible planning algorithms that, no matter the drivetrain. However, it is useful to consider this situation since holonomic grid search algorithms are popular.

Further, it is a common approach for local trajectory algorithms to leverage a rolling environmental model with the robot at its center. The predictive collision detection algorithm will check for collisions some N seconds in the future given the robot’s current commanded speed. If a near-future collision is detected, the robot is stopped. When a robot is traveling at high speeds, this may actually lapse outside of the bounds of a rolling environmental model if not carefully tuned. The implementation offers protections from such situations and issues stern recommendations to prevent the serious safety violation.

Users are able to tune all of the parameters related to PP, APP, and RPP. In particular, the heuristics have parameters r_{min} , α , and d_{prox} . It

is possible for these to be poorly tuned such that a robot would drop to an impractically low speed when in close proximity to an obstacle or in a turn. To prevent the robot from traveling *too* slow, a minimum speed threshold is exposed such that the robot’s speed will never be set below a parameterized value. Note: the parameters and speeds in the experiment were set up such that this did not impact our experimental results in Section 5.

An open-source community contributed feature made available is support for reversing - not only driving forward. When a path contains a cusp or discontinuity in direction, the controllers will reverse from forward to back and vis-a-vis. When paired with a planner that can generate such paths, such as the Hybrid-A* planner in Nav2, the robot can track more complex paths.

5 Experiments and Analysis

This section describes the experiments to demonstrate the benefits of our contribution compared to other existing Pure Pursuit variants. One experiment was conducted in simulation and three others were conducted with a physical Tiago robot (Figure 2). Tiago is used in industrial and service applications and the test environment is shown in Figure 7, corresponding to a university campus building.

In all three hardware experiments below, the same navigation configurations are utilized. The maximum linear speed v_{max} is set to $0.8m/s$, the maximum acceleration a_{max} is $0.2m/s^2$, and the maximum angular speed ω_{max} is $3.2rad/s^2$. The lookahead distance L_t is set between $0.25m$ and $1.2m$ and lookahead time is $1.0s$ for Adaptive and Regulated Pure Pursuit; while Pure Pursuit uses the maximum of $1.2m$. In all four experiments, no replanning was utilized such that the distance to path data is regular and can be meaningfully analyzed. SLAM Toolbox was used to generate the 2D map and AMCL was used to localize within it [5, 14].

5.1 Path Tracking Experiment

A simulation experiment was conducted to analyze the improvements Regulated Pure Pursuit can offer during sharp turns in ideal conditions. This simulation was conducted using the Turtlebot 3 robot, the Gazebo simulator, and an empty

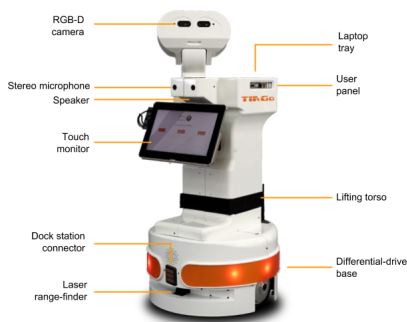


Fig. 2 实验中使用的 Pal Robotics Tiago [12]。

这项工作专门针对工业和服务机器人，其中许多是差速驱动的。虽然这对 Pure Pursuit 来说不是技术问题，但许多差动驱动机器人将使用基于完整搜索的路径规划器（例如 Dijkstra 或 A*），而不是阿克曼车辆经常使用的运动学上可行的规划器[13]。这是受监管的纯粹追踪实施所关注的问题，因为路径的相对航向可能与机器人的起始航向不一致。在这种情况下，机器人能够在开始跟踪完整路径之前旋转到粗略的起始航向是有益的。此外，当机器人部署在方向敏感的应用中时，当机器人处于平移目标容差内时旋转到最终航向非常有用。当使用可行的规划算法时，无论传动系统如何，这些都不是问题。然而，考虑这种情况是有用的，因为完整网格搜索算法很流行。

此外，局部轨迹算法的常见方法是利用以机器人为中心的滚动环境模型。考虑到机器人当前的命令速度，预测碰撞检测算法将在未来的 N 秒内检查碰撞情况。如果检测到近期发生碰撞，机器人就会停止。当机器人高速行驶时，如果不仔细调整，它实际上可能会超出滚动环境模型的范围。该实施提供了针对此类情况的保护，并提出了严厉的建议，以防止严重的安全违规行为。

用户可以调整与 PP、APP 和 RPP 相关的所有参数。特别地，启发式具有参数 r_{min} 、 α 和 d_{prox} 。它

这些可能会调整得不好，导致机器人在靠近障碍物或转弯时会降至不切实际的低速。为了防止机器人缓慢行进 *too*，设置了最小速度阈值，这样机器人的速度永远不会低于参数化值。注意：实验中的参数和速度的设置不会影响第 5 节中的实验结果。

开源社区贡献的一项功能是支持倒车，而不仅仅是前进。当路径包含尖点或方向不连续时，控制器将从前向后以及相对过孔反向。当与可以生成此类路径的规划器（例如 Nav2 中的 Hybrid-A* 规划器）配合使用时，机器人可以跟踪更复杂的路径。

5 Experiments and Analysis

本节描述了一些实验，以证明我们的贡献与其他现有的 Pure Pursuit 变体相比的好处。一项实验是通过模拟进行的，另外三项实验是使用物理 Tiago 机器人进行的（图 2）。Tiago 用于工业和服务应用，测试环境如图 7 所示，对应于大学校园建筑。

在下面的所有三个硬件实验中，都使用相同的导航配置。最大线速度 v_{max} 设置为 $0.8m/s$ ，最大加速度 a_{max} 为 $0.2m/s^2$ ，最大角速度 ω_{max} 为 $3.2 rad/s^2$ 。对于自适应和调节纯追踪，前瞻距离 L_t 设置在 $0.25m$ 和 $1.2m$ 之间，前瞻时间为 $1.0s$ ；而 Pure Pursuit 使用最大值 $1.2m$ 。在所有四个实验中，没有使用重新规划，因此到路径数据的距离是规则的并且可以进行有意义的分析。SLAM Toolbox 用于生成 2D 地图，并使用 AMCL 在其中进行定位 [5, 14]。

5.1 Path Tracking Experiment

我们进行了一项模拟实验，以分析“Regulated Pure Pursuit”在理想条件下急转弯时可以提供的改进。该模拟是使用 Turtlebot 3 机器人、Gazebo 模拟器和一个空的

environment with ground truth information made available to remove the contribution of odometric and localization error. The robot followed the reference step function path indicated by the red piece-wise line in Figures 3 and 4 using the three variants of Pure Pursuit.

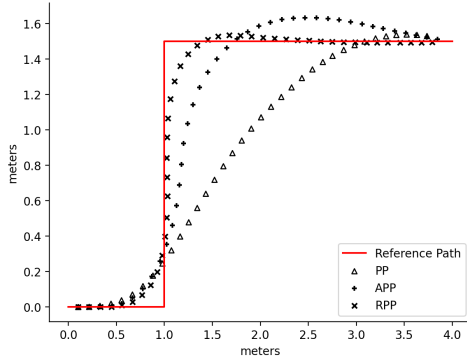


Fig. 3 Comparison of variants of Pure Pursuit.

Figure 3 displays the comparison with the different methods. The same parameters were utilized in this experiment as the hardware experiments, with the exception of the desired linear velocity, set to 1.0m/s and the curvature heuristic minimum radius, set to 1.5m . The overshoot observed during sharp turns was notably lower in Regulated Pure Pursuit due to the controlled slow down during sharp changes in curvature allowing the robot to track the path more finely. The slow down also triggers the use of a closer adaptive lookahead point further increasing local stability. The mean tracking error from the reference path of Regulated Pure Pursuit was merely 0.03m . The others experienced over an order of magnitude difference at 0.10m and 0.19m for APP and PP, respectively. While the main contribution of this work is *not* an improvement on path tracking, it is a convenient emergent property to further ensure safety. This experiment showcases that the algorithm can follow a computed collision-free path in confined spaces more closely and thusly more safely.

Using the ideal environment, the effects of varying the curvature heuristic’s minimum radius were also studied. From Figure 4, it can be observed that larger values of r_{min} allow a robot to follow the reference path more closely. This is

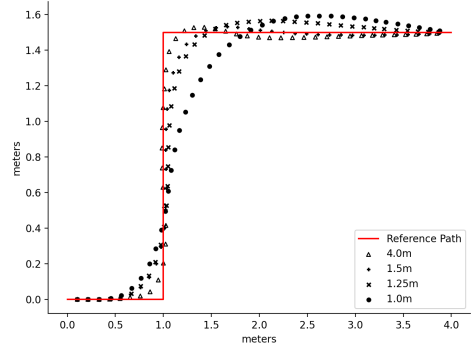


Fig. 4 Effects of varying r_{min} in Regulated Pure Pursuit.

due to the curvature triggering reactions quickly on approach to the discontinuity. When r_{min} was increased from 1.0m to 1.5m , the mean tracking error decreases from 0.8m to 0.3m . This reduction in tracking error does come at a cost: a reduction in the speed resulting in longer navigation times. For this experiment, when $r_{min} > 1.5\text{m}$, the improvement on tracking error became insignificant, but navigation times begin to spike as the speed during turns trends towards the minimum allowed speed.

5.2 Blind Turning Experiment

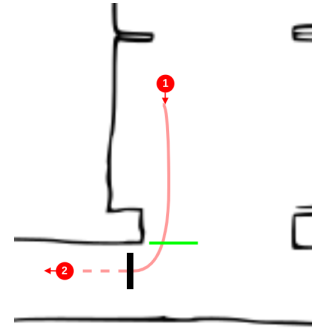


Fig. 5 A blind corner experiment with an obstacle just out of view. When the robot cross the green line, the obstacle is set.

This experiment evaluates the three algorithms’ performance when the robot encounters an unexpected obstacle that blocks its path around a blind corner. The experiment consists of the robot making a sharp 90-degree turn (Figure 5) behind which there is an obstacle. The obstacle is such that the robot cannot observe it before beginning

具有地面实况信息的环境可用于消除里程计和定位误差的影响。机器人使用 Pure Pursuit 的三种变体，遵循图 3 和图 4 中红色分段线所示的参考阶跃函数路径。

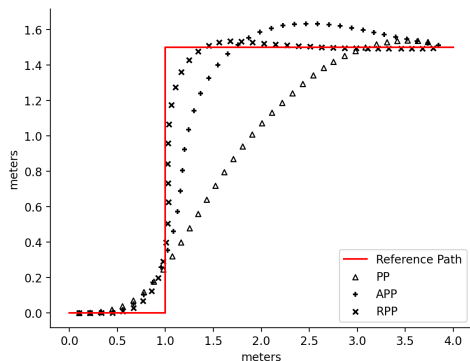


Fig. 3 纯粹追求的变体比较。

图 3 显示了不同方法的比较。该实验使用与硬件实验相同的参数，除了所需的线速度设置为 1.0 m/s 和曲率启发式最小半径设置为 1.5 m 之外。在“调节纯粹追踪”中，急转弯期间观察到的超调明显较低，因为曲率急剧变化期间的受控减速使机器人能够更精细地跟踪路径。减速还触发了更接近的自适应前瞻点的使用，进一步提高了局部稳定性。调节纯粹追踪的参考路径的平均跟踪误差仅为 0.03 m 。其他人的 APP 和 PP 分别在 0.1 m 和 0.19 m 处经历了一个数量级的差异。虽然这项工作的主要贡献是 *not* 对路径跟踪的改进，但它是进一步确保安全的方便的紧急属性。该实验表明，该算法可以在有限空间内更紧密地遵循计算出的无碰撞路径，从而更安全。

使用理想环境，还研究了改变曲率启发式最小半径的影响。从图 4 中可以看出，较大的 r_{min} 值允许机器人更紧密地遵循参考路径。这是

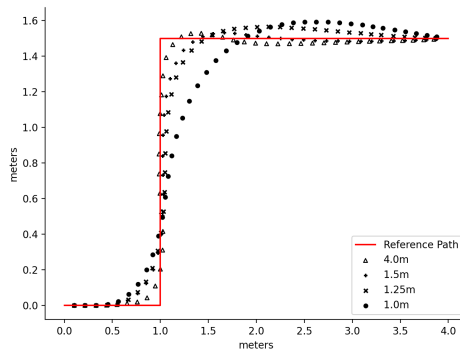


Fig. 4 调节纯粹追逐中不同 r_{min} 的效果。

由于曲率在接近不连续性时会快速触发反应。当 r_{min} 从 1.0 m 增加到 1.5 m 时，平均跟踪误差从 0.8 m 减少到 0.3 m 。跟踪误差的减少确实是有代价的：速度的降低导致导航时间更长。对于此实验，当 $r_{min} > 1.5\text{ m}$ 时，跟踪误差的改善变得微不足道，但随着转弯期间的速度趋向于最小允许速度，导航时间开始激增。

5.2 Blind Turning Experiment

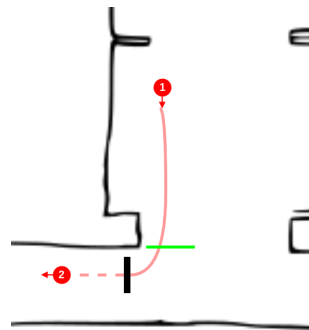


Fig. 5 盲角实验，障碍物刚好在视野之外。当机器人越过绿线时，障碍物就设置好了。

该实验评估了当机器人遇到意外障碍物阻挡其绕过盲角的路径时这三种算法的性能。该实验包括机器人急转 90° (图 5)，后面有一个障碍物。障碍物使得机器人在开始之前无法观察到它

to execute the curve and has minimal time to react, placed after the robot crosses the green line.

For each algorithm, the experiment is repeated ten times. The purpose of this experiment is to test the reaction of each algorithm in a common but particularly dangerous situation. While PP and APP do not themselves contain collision detection capabilities, for a fair comparison, each are evaluated using the collision checking method described in Section 3.

Table 1 Result of the Blind Turning Experiment.

	PP	APP	RPP
Avg Stopped Distance (m)	0.15	0.16	0.24

Table 1 shows the results of this experiment; in all cases, none of the methods resulted in a collision. This shows that the collision detection features of RPP (applied to all methods) are sufficient to prevent blind collisions at service robot speeds. A similar experiment without RPP’s collision detection yielded collisions in nearly all instances. The RPP algorithm displayed a 33% increase in average stopped distance relative to APP. APP and PP, both which do not change their speeds in turns, had expectedly similar behavior. This is an improvement in reaction time for safe navigation in environments with many dynamic agents potentially cornering at the same time. While the 8cm increase does not seem like much, with many robots navigating independently for their own tasks, this additional precaution is the difference between robots scraping by and colliding with permanent damage. This highlights that RPP’s slowing of the robot in blind turns helps to reduce risk in partially observable settings.

5.3 Confined Corridor Experiment

The confined corridor experiment was designed to evaluate how well the proposed approach improves performance in narrow spaces at high speeds when making continuous changes in direction. Figure 6 shows the test environment map, creating a "slalom" pattern, starting from the left. The total width of the corridor is 1.5 meters and the obstacles each have dimensions of roughly 0.7 m. Each algorithm was tested five times and the data from this experiment is shown in Table 2.

Table 2 Result of the confined corridor experiment.

	PP	APP	RPP
Time (s)	13.0	12.7	13.3
Distance (m)	9.04	9.54	9.63
Collisions	0	0	0
Average Speed (m/s)	0.679	0.736	0.682
Average Distance Obstacle (m)	0.662	0.681	0.683
Average Distance to Path (m)	0.100	0.059	0.052

Figure 6 displays the robot position data overlaying the trajectory each algorithm was asked to follow. The most telling region of this experiment to analyze is the final and sharpest turn. RPP’s proximity and curvature heuristics contributed in this location and resulted in a reduced speed and 14% less path tracking error than APP. The other algorithms had path undershoot due to the sharp turn and came into much closer proximity to the obstacle and wouldn’t be considered functionally safe. This undershoot was corrected in RPP by reacting to the increased curvature partially due to the deviation from the path, compounded by its approaching proximity to the obstacle, slowing the robot down. This allowed RPP to better recover from a minor deviation off the path and navigated safer distance away from the obstacle. The impact of the regulation heuristics can be seen in blue as the robot weaves around the final obstacle after quickly changing direction.

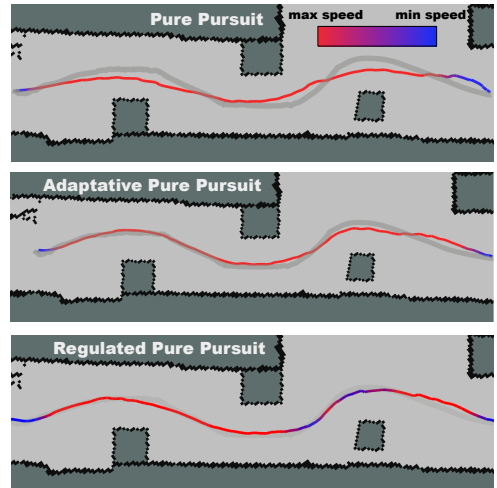


Fig. 6 Calculated path in gray and the robot’s traveled path, colored by speed, during the corridor experiment.

The distance traveled by PP is lower than both variants due to the short cutting it displayed

执行曲线并有最短的反应时间，放置在机器人越过绿线之后。

对于每个算法，实验重复十次。这个实验的目的是测试每种算法在常见但特别危险的情况下的反应。虽然 PP 和 APP 本身不包含碰撞检测功能，但为了公平比较，每个都使用第 3 节中描述的碰撞检查方法进行评估。

Table 1 盲转实验的结果。

	PP	APP	RPP
Avg Stopped Distance (m)	0.15	0.16	0.24

表1为本次实验的结果；在所有情况下，这些方法都没有导致碰撞。这表明 RPP 的碰撞检测功能（适用于所有方法）足以防止服务机器人速度下的盲目碰撞。没有 RPP 碰撞检测的类似实验几乎在所有情况下都会产生碰撞。相对于 APP，RPP 算法显示平均停止距离增加了 33%。APP 和 PP 都不会轮流改变速度，但它们的行为与预期类似。这改善了在许多动态代理可能同时转弯的环境中安全导航的反应时间。虽然 8cm 的增加看起来并不多，因为许多机器人独立导航以完成自己的任务，但这种额外的预防措施是机器人刮擦和碰撞造成永久性损坏的区别。这凸显了 RPP 在盲转弯时减慢机器人速度有助于降低部分可观察环境中的风险。

5.3 Confined Corridor Experiment

受限走廊实验旨在评估所提出的方法在连续改变方向时在高速狭窄空间中提高性能的效果。图 6 显示了测试环境图，从左侧开始创建了“回转”模式。走廊总宽度为 1.5 米，每个障碍物的尺寸约为 0.7 m。每个算法都经过了五次测试，实验数据如表2所示。

Table 2 受限走廊实验的结果。

	PP	APP	RPP
Time (s)	13.0	12.7	13.3
Distance (m)	9.04	9.54	9.63
Collisions	0	0	0
Average Speed (m/s)	0.679	0.736	0.682
Average Distance Obstacle (m)	0.662	0.681	0.683
Average Distance to Path (m)	0.100	0.059	0.052

图 6 显示了机器人位置数据，覆盖了每个算法要求遵循的轨迹。该实验中最需要分析的区域是最后也是最急的转弯。RPP 的邻近度和曲率启发式算法在此位置上发挥了作用，与 APP 相比，速度降低，路径跟踪误差减少 14%。其他算法由于急转弯而出现路径下冲，并且距离障碍物更近，因此不被认为是功能安全的。这种下冲在 RPP 中得到了纠正，部分是由于偏离路径而导致的曲率增加，再加上它接近障碍物，从而减慢了机器人的速度。这使得 RPP 能够更好地从偏离路径的微小偏差中恢复过来，并导航到远离障碍物的更安全距离。当机器人在快速改变方向后绕着最后一个障碍物移动时，可以用蓝色看到调节启发式的影响。

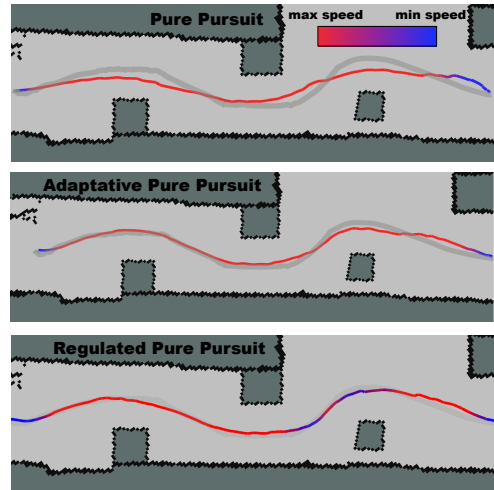


Fig. 6 在走廊实验期间，计算出的路径为灰色，机器人的行进路径按速度着色。

PP 行驶的距离比这两种变体都要短，因为它显示的距离较短

throughout the trajectory in Figure 6. This habitual undershoot also led to an decreased average distance to an obstacle - passing much closer to potential collisions.

5.4 Full-system experiment

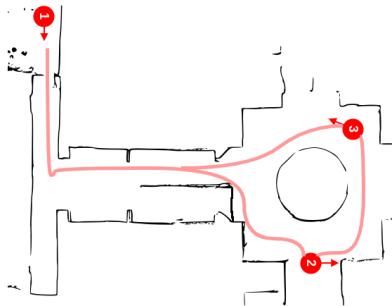


Fig. 7 Itinerary for full-system experiment. The total length is 70 meters.

In the final experiment, the robot will follow a route through the three control points shown in Figure 7 in a campus building. The route contains collision-free open space, confined halls, and blind sharp turn of approximately 70 m. The robot repeats the path three times with each of the algorithms, the data is provided in Table 3. The purpose of this experiment is to compare the RPP algorithm’s general behaviors at a system-level to the existing variants.

The data indicates little difference in overall system-level behavior - the average time to navigate was between 85-95s. Consistent with the previous experiment, PP does display slightly shorter distance traveled and an increased tracking error - corresponding to path short-cutting.

Table 3 Result of the full-system experiment.

	PP	APP	RPP
Time (s)	85.6	94.3	88.6
Distance (m)	58.24	59.26	58.64
Collisions	0	0	0
Average Speed (m/s)	0.661	0.675	0.646
Min Distance to Obstacle (m)	1.139	1.133	1.135
Average Distance to Path (m)	0.062	0.049	0.043

The most compelling attribute of this experiment is the lack of particularly unique outcomes between the algorithms. Although RPP slows the

robot in narrow corridors and while completing sharp turns, surprisingly, these maneuvers did not meaningfully impact the high-level navigation metrics. Time to task completion and average speeds were consistent across all three algorithms when considering the standard deviations between the trials. The system-level performance of RPP is so similar to that of APP, we conclude that the additional benefits of RPP come at little disadvantage to a system designer. In fact, it is possible for a solution architect to increase the maximum speed of the robot modestly when using RPP due to its slowing in turns and in proximity to obstacles (the common limiting factors of speed in robotics applications). This meaningfully improves the overall efficiency of a robot system while securing higher-quality safety features.²

6 Limitations

Regulated Pure Pursuit, and other Pure Pursuit variants, suffer from a lack of modeling of the vehicle. As this class of technique is purely geometric based on the path, it will not consider dynamic constraints while changing velocities to track the path. Further, the global path must be feasibly drivable for a given robot platform since these methods compute velocities in the absence of kinematic limitations. For differential-drive robots, this may be any path. However for Ackermann steering robots, this path must already be drivable considering the kinematic limitations on the minimum possible turning radius.

Regulated Pure Pursuit also continues to short-cut sections of paths with high curvature turns, though to a much reduced degree than PP or APP. The work introduced in [?] provides an alternative method for selecting the lookahead point with the goal of more accurately tracking the reference path. While this work also introduces a velocity heuristic that is not suitable for our class of applications, a variation on the method for selecting lookahead points has merit to be applied in mobile robotics.

²A video with the experiments with the real robot can be found at <https://youtu.be/LQAzzJ8GmS0>

图 6 中的整个轨迹。这种习惯性的下冲还导致到障碍物的平均距离缩短——距离潜在碰撞更近。

5.4 Full-system experiment

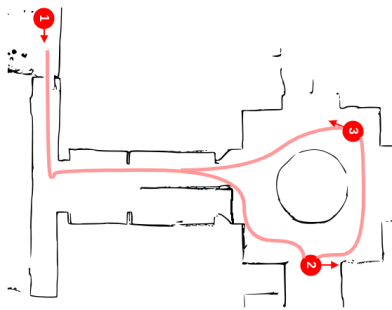


Fig. 7 全系统实验行程。总长度为70米。

在最终实验中，机器人将沿着校园建筑中如图 7 所示的三个控制点的路线行驶。该路线包含无碰撞开放空间、密闭大厅和约 70 m 的盲急转弯。机器人使用每种算法重复该路径 3 次，数据如表 3 所示。本实验的目的是将 RPP 算法在系统级的一般行为与现有变体进行比较。

数据表明整体系统级行为差异不大——平均导航时间在 85-95 秒之间。与之前的实验一致，PP 确实显示出行驶距离稍短且跟踪误差增加 - 对应于路径捷径。

Table 3 全系统实验的结果。

	PP	APP	RPP
Time (s)	85.6	94.3	88.6
Distance (m)	58.24	59.26	58.64
Collisions	0	0	0
Average Speed (m/s)	0.661	0.675	0.646
Min Distance to Obstacle (m)	1.139	1.133	1.135
Average Distance to Path (m)	0.062	0.049	0.043

该实验最引人注目的属性是算法之间缺乏特别独特的结果。尽管 RPP 减缓了

令人惊讶的是，当机器人在狭窄的走廊中完成急转弯时，这些动作并没有对高级导航指标产生有意义的影响。当考虑试验之间的标准偏差时，所有三种算法的任务完成时间和平均速度都是一致的。RPP 的系统级性能与 APP 非常相似，因此我们得出结论，RPP 的额外优势对系统设计人员来说几乎没有什么缺点。事实上，解决方案架构师在使用 RPP 时可以适度提高机器人的最大速度，因为机器人轮流和靠近障碍物时速度会减慢（机器人应用中常见的速度限制因素）。这有意义地提高了机器人系统的整体效率，同时确保了更高质量的安全功能。²。

6 Limitations

Regulated Pure Pursuit 和其他 Pure Pursuit 变体都缺乏车辆建模。由于此类技术是纯粹基于路径的几何技术，因此在改变速度来跟踪路径时不会考虑动态约束。此外，对于给定的机器人平台，全局路径必须是可行的，因为这些方法在没有运动学限制的情况下计算速度。对于差动驱动机器人，这可以是任何路径。然而，对于阿克曼转向机器人来说，考虑到最小可能转弯半径的运动学限制，这条路径必须已经可以行驶。

Regulated Pure Pursuit 还继续缩短具有高曲率转弯的路径部分，但程度比 PP 或 APP 低得多。[?]中介绍的工作提供了一种选择前瞻点的替代方法，其目标是更准确地跟踪参考路径。虽然这项工作还引入了不适合我们的应用类别的速度启发式方法，但选择前瞻点的方法的变体值得应用于移动机器人技术。

²A video with the experiments with the real robot can be found at <https://youtu.be/LQAzzJ8GmS0>

7 Conclusion

Regulated Pure Pursuit builds incrementally on Adaptive Pure Pursuit with a focus on service robots. This method contains a schema for selecting velocities that improves functional safety for real-world deployed robot applications.

Our demonstrations on an industrial robot exhibited improved safety around blind turns and in confined settings without significant system-level changes. A high-quality implementation of Regulated Pure Pursuit is freely available at <https://github.com/ros-planning/navigation2> and is in use on robots today.

Conflict of Interest and Acknowledge

The authors have no conflicts of interest to declare that are relevant to the content of this article.

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7 Conclusion

Regulated Pure Pursuit 在 Adaptive Pure Pursuit 的基础上逐步构建, 重点关注服务机器人。该方法包含一个用于选择速度的模式, 可提高实际部署的机器人应用程序的功能安全性。

我们对工业机器人的演示表明, 在盲转弯和受限环境中的安全性有所提高, 而无需进行重大的系统级更改。Regulated Pure Pursuit 的高质量实现可在 <https://github.com/ros-planning/navigation2> 免费获得, 并且现已在机器人上使用。

Conflict of Interest and Acknowledge

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